The Continuous-Time Fourier Transform (CTFT)

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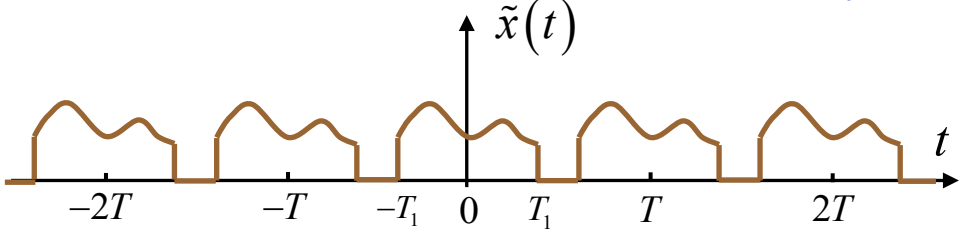
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- Fourier Representation of Signals
  - The Fourier method is the most extensively applied signal processing tool.
  - The transform output lends itself to easy interpretation and manipulation, and leads to the concept of frequency analysis
  - The power of the Fourier transform in signal is its ability to reveals spectral structures that may be used to characterize a signal.
  - The applications of the Fourier transform include filtering, telecommunication, music processing, pitch modification, signal coding, pattern identification, speech recognition, image processing, spectral analysis in astrophysics, radar signal processing.

From FS to FT

- Periodic signal can be represented by the linear combination of harmonically related complex exponentials



$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \qquad \omega_0 = \frac{2\pi}{T}$$

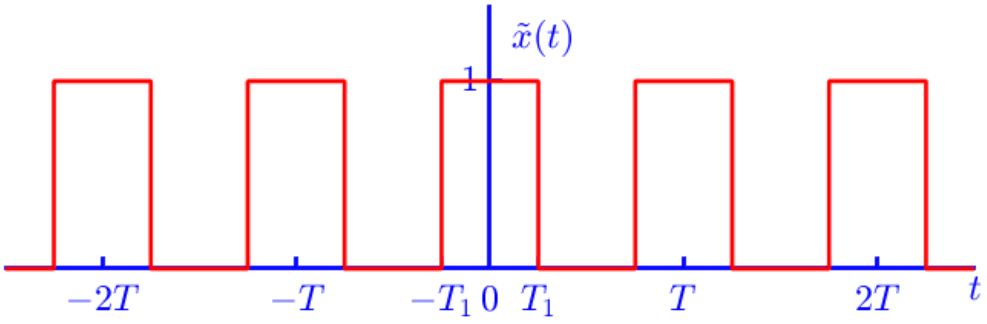
  - The harmonic $e^{jk\omega_0 t}$ passing through LTI yields

$$y(t) = h(t) * e^{j\omega_0 t} = e^{j\omega_0 t} \int_{-\infty}^{+\infty} h(\tau) e^{-j\omega_0 \tau} d\tau = e^{j\omega_0 t} H(j\omega_0)$$

- Hence $\tilde{x}(t) * h(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$

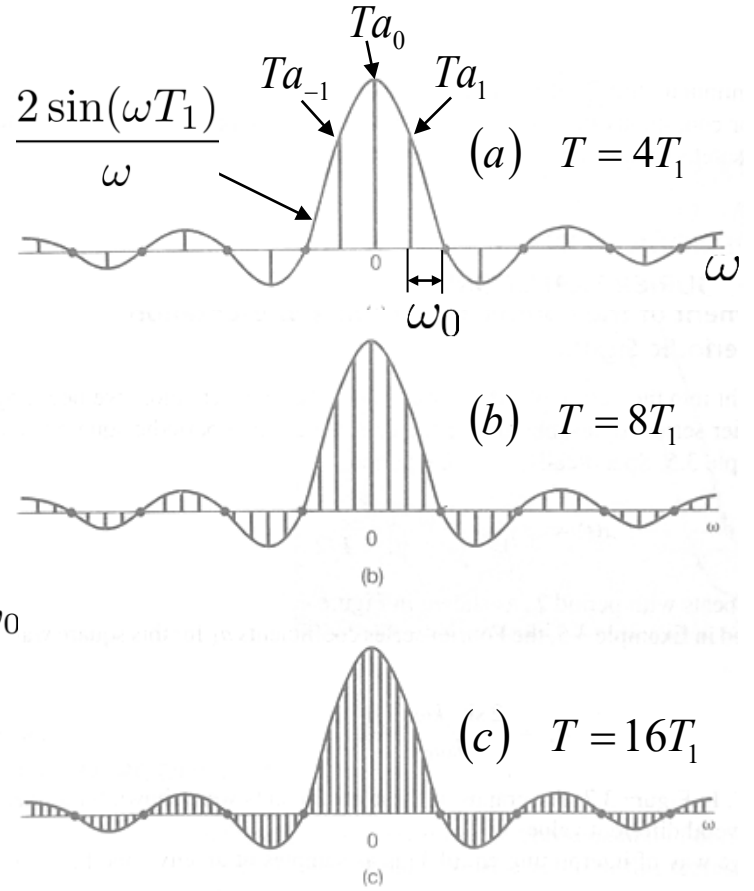
1. We will extend FS to aperiodic, finite energy signals. The resulting representation is referred to as Fourier transform (FT).
2. Since an aperiodic signal can be viewed as a periodic signal with an infinite period, we can deduce FT from FS.

Representation of Aperiodic Signals



$$\tilde{x}(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

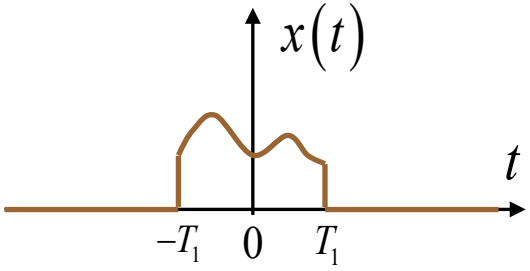
$$a_k = \frac{1}{T}\frac{2\sin(k\omega_0 T_1)}{k\omega_0}, \quad Ta_k = \frac{2\sin(\omega T_1)}{\omega}\Big|_{\omega = k\omega_0}$$



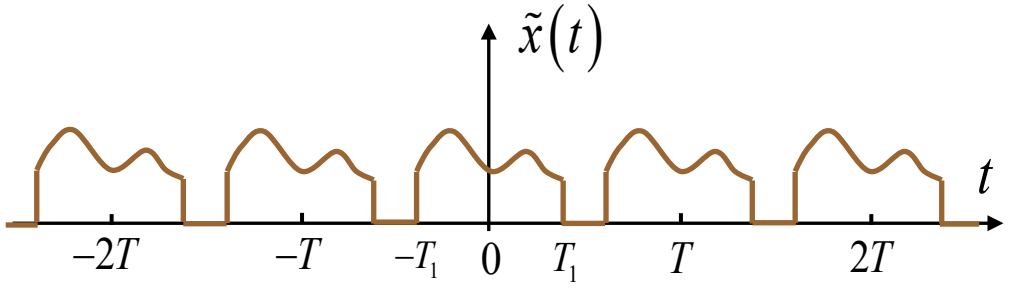
fix T_1 , increase T

- Ta_k is the equally spaced samples of the envelope
- With the increasing of T, the sampling distance reduces
- As T goes to infinity, $\tilde{x}(t)$ becomes aperiodic, and Ta_k is exactly the envelope

⇨ Not FS



aperiodic signal



periodic signal

- For the above periodical signal, there is

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} \tilde{x}(t)e^{-jk\omega_0 t}dt \qquad \tilde{x}(t) = \sum_{-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

- From the relationship between $\tilde{x}(t)$ and $x(t)$, there is

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} \tilde{x}(t)e^{-jk\omega_0 t}dt = \frac{1}{T}\int_{-\infty}^{\infty} x(t)e^{-jk\omega_0 t}dt$$

- By observation, if we define the envelop of Ta_k

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

→ Fourier Transform (Spectrum)

  - then $a_k = \frac{1}{T}X(jk\omega_0)$ Not the direct sampling $X(j\omega)$, T must be there
  - Meanwhile

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty} X(jk\omega_0)e^{jk\omega_0 t}\omega_0$$

$$\frac{1}{T} = \frac{\omega_0}{2\pi}$$

  - From $x(t) = \lim_{\omega_0 \to 0} \tilde{x}(t)$, we compute

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$$

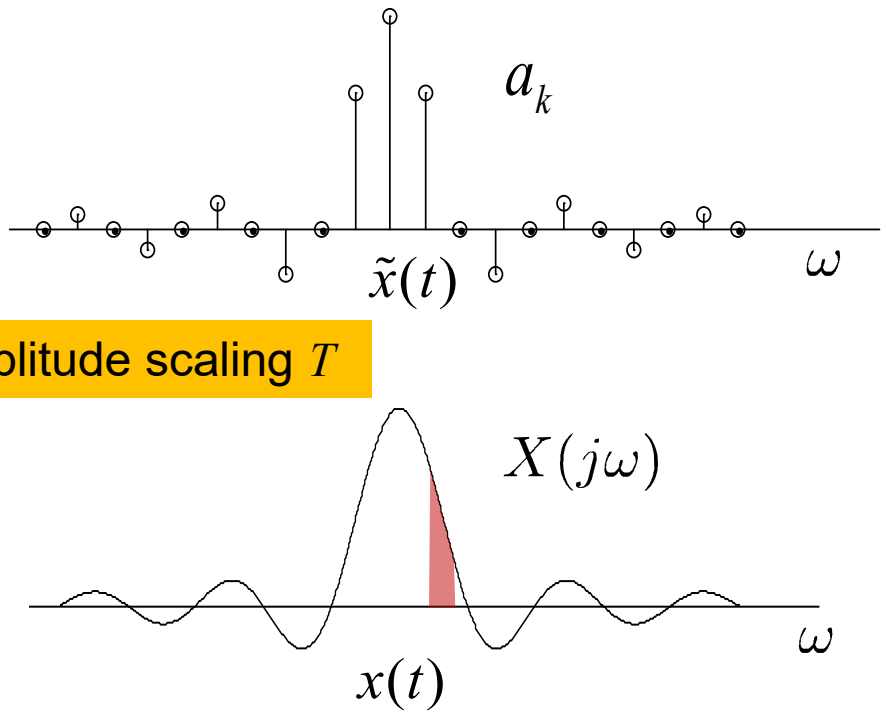
→ Inverse Fourier Transform

Aperiodic signal can be represented by the combination of infinite number of complex harmonics $e^{j\omega t}$ with continuous frequency and amplitude $X(j\omega)d\omega/2\pi$

- Clearly the FS and the FT are related as

$$a_k = \frac{1}{T} X(k\omega_0)$$

- The spectrum of periodic signal is the sampling of the spectrum of the corresponding aperiodic, with a scaling *1/T*
- The spectrum of aperiodic signal is the envelope of the spectrum of the corresponding periodic, with a scaling T



The portion in $x(t)$ that is within frequency $[\omega_1,\ \omega_2]$

$$\frac{1}{2\pi}\int_{\omega_1}^{\omega_2} X(j\omega)e^{j\omega t}d\omega$$

Physical Explanation of FT

- Check which time domain signal can have FT $2\pi\delta(\omega-\omega_0)$

$$\frac{1}{2\pi}\int_{-\infty}^{+\infty} 2\pi\delta(\omega-\omega_0)e^{j\omega t}d\omega = e^{j\omega_0 t}$$

Here, ω_0 is any constant

  - Namely

$$\int_{-\infty}^{+\infty} e^{j\omega_0 t}e^{-j\omega t}dt = 2\pi\delta(\omega-\omega_0)$$

Complex exponentials with different frequencies are orthogonal

  - Next check

$$X(j\omega_0)=\int_{-\infty}^{+\infty}x(t)e^{-j\omega_0 t}dt=\frac{1}{2\pi}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}X(j\omega)e^{j(\omega-\omega_0)t}d\omega dt$$

$$=\int_{-\infty}^{+\infty}X(j\omega)\frac{1}{2\pi}\int_{-\infty}^{+\infty}e^{j(\omega-\omega_0)t}dtd\omega=\int_{-\infty}^{+\infty}X(j\omega)\delta(\omega-\omega_0)d\omega$$

FT could extract the frequency component from the original spectrum

Convergence of Fourier Transform

- Convergence of Fourier Transform

$$\hat{x}(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega \doteq x(t)$$

  - From the derivation of FT, the same conditions for FS could also be applied for FT

(a) If $x(t)$ has finite energy $\int_{-\infty}^{\infty} |x(t)|^2 dt < +\infty$

(b) Dirichlet conditions

1) Absolutely integrable $\int_{-\infty}^{\infty} |x(t)|dt < \infty$

2) $x(t)$ is of bounded variation, i.e., have finite number of maxima and minima within any finite interval

3) $x(t)$ have finite number of discontinuities within any finite interval. Furthermore, each of these discontinuities is finite.

- These two conditions are not equal
  - For square integrable one, there can be different individual values.
  - For Dirichlet conditions, the value at discontinuity is the average of values on both sides
  - Example: $\sin t/t$ is not absolutely integrable but is square integrable. (just sinc function)

Gibbs phenomenon appears

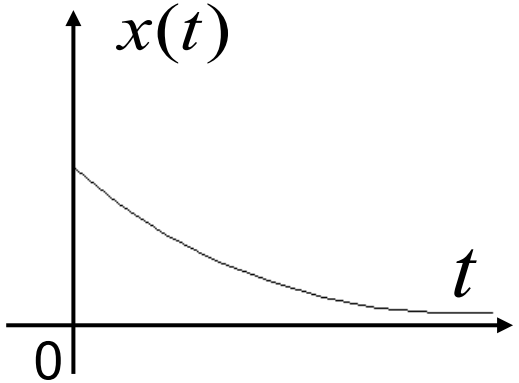
- These two conditions are only sufficient
  - We will see that the periodic signals do not satisfy either condition, but it can have FT if impulse functions are allowed. Then we could also represent periodical signal with FT, and thus get a unified framework for FT and FS.

Remember x(t)=1

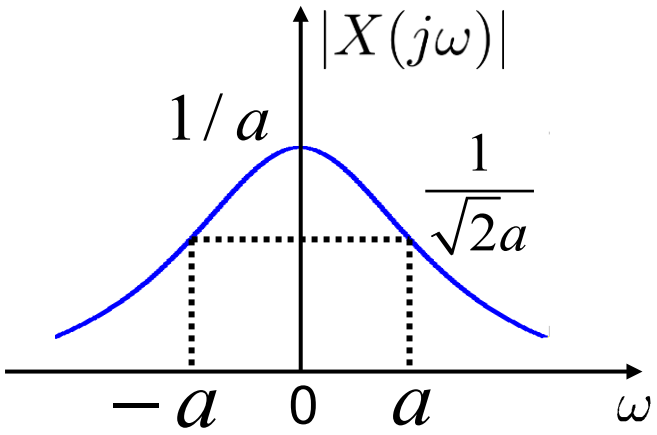
Commonly Used CTFT

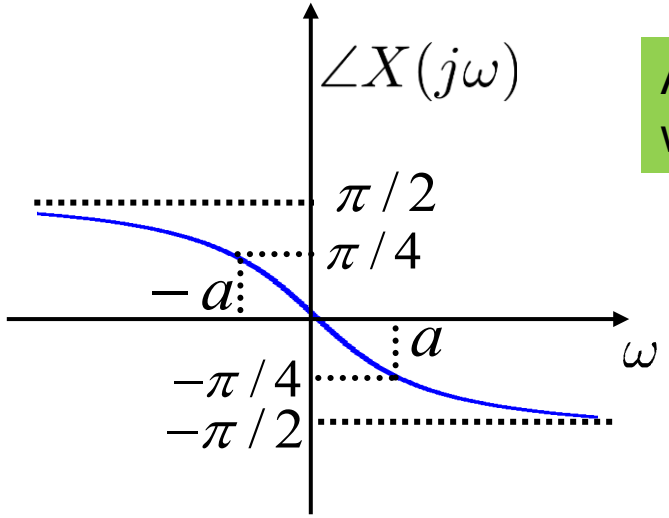
- Example 1
  - $x(t) = e^{-at}u(t), \quad a > 0$

$$X(j\omega) = \int_0^\infty e^{-at}e^{-j\omega t}dt = \frac{1}{a + j\omega}$$



$$|X(j\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}}, \qquad \angle X(j\omega) = -\tan^{-1}\left(\frac{\omega}{a}\right) = -\angle(a + j\omega)$$





Actually, $\Re\{a\} > 0$ will do

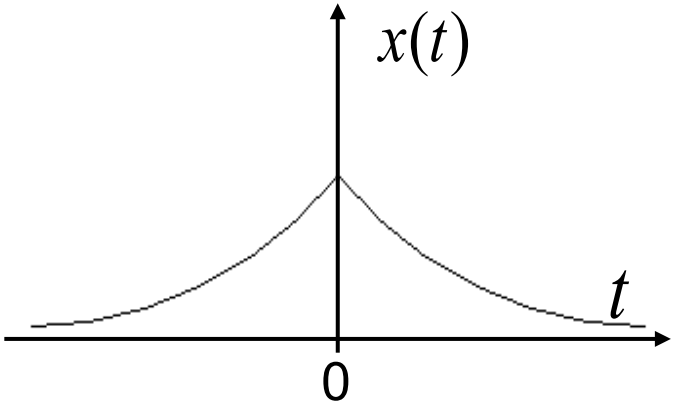
Could we get the FT of $u(t)$ as $a \to 0$? ?

- Example 2

$$x(t) = e^{-a|t|}, \quad a > 0$$

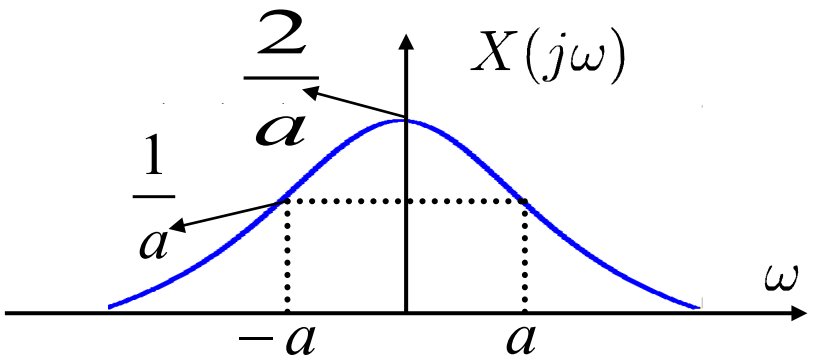
$$X(j\omega) = \int_{-\infty}^{0} e^{at} e^{-j\omega t} dt + \int_{0}^{+\infty} e^{-at} e^{-j\omega t} dt$$

$$= \frac{1}{a - j\omega} + \frac{1}{a + j\omega} = \frac{2a}{a^2 + \omega^2}$$



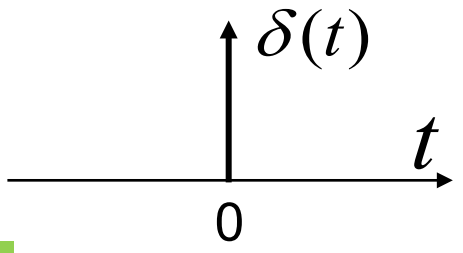
$$|X(j\omega)| = X(j\omega)$$

$$\angle X(j\omega) = 0$$

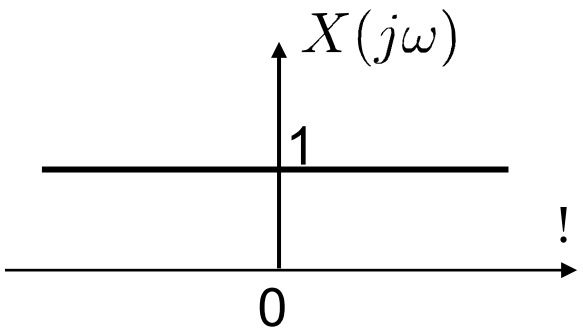


- Example 3: Unit impulse

$$x(t) = \delta(t) \qquad X(j\omega) = 1$$

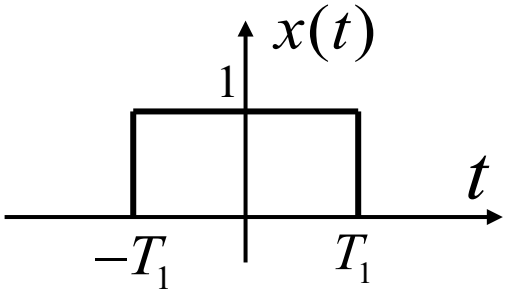


$\delta(t)$ includes all frequency components, and the amplitude/phase of each component is the same. This is why the impulse response of LTI system $h(t)$ can completely characterize the system



- Example 4: Rectangular pulse

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases}$$

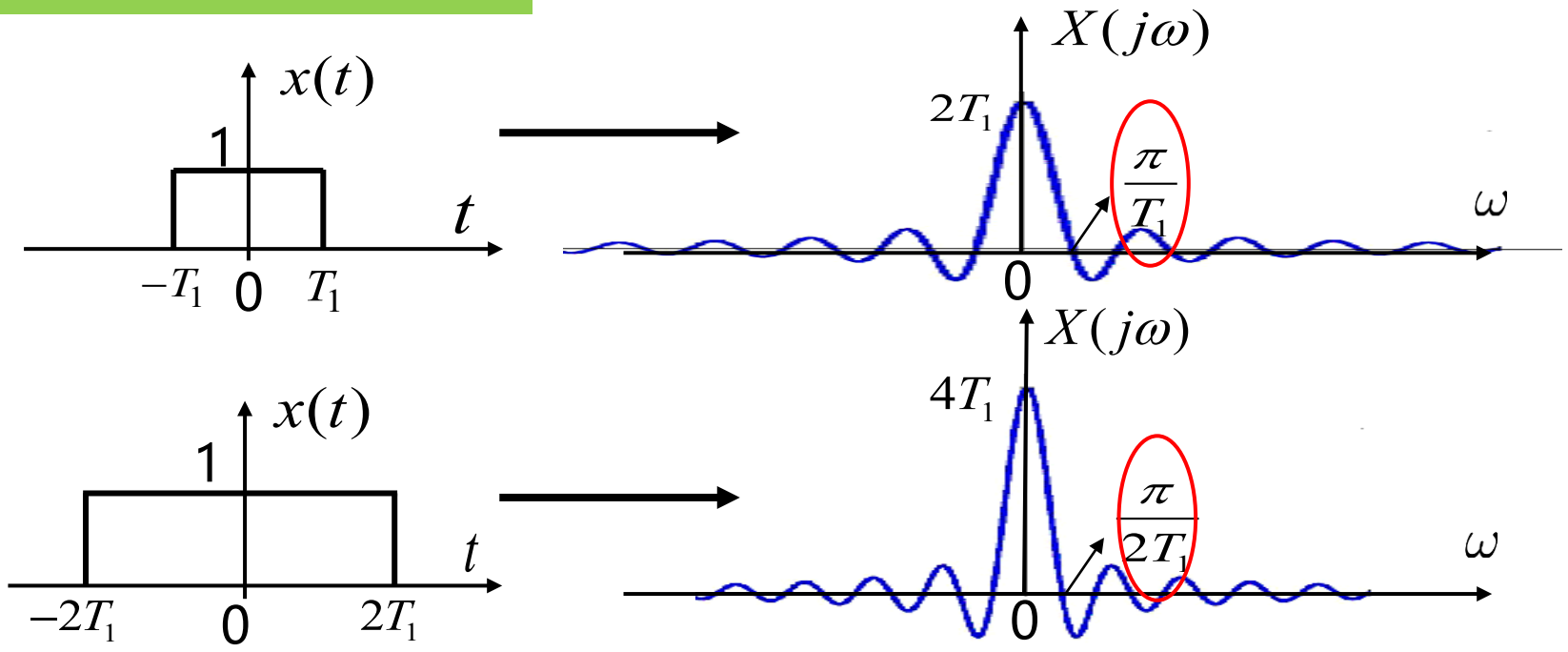


$$X(j\omega) = \int_{-T_1}^{T_1} e^{-j\omega t} dt = -\frac{1}{j\omega} e^{-j\omega t} \Big|_{-T_1}^{T_1} = \frac{2\sin \omega T_1}{\omega}$$

$$X(j\omega) = \frac{2\sin\omega T_1}{\omega} = 2T_1\,\mathrm{sinc}\left(\frac{\omega T_1}{\pi}\right)$$

Remember what is FS for periodic rectangular pulse?

Check different time duration



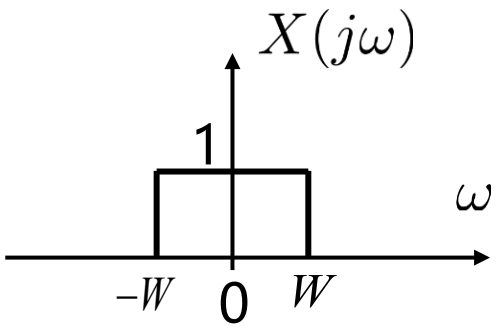
Inverse relationship between time and frequency domain!

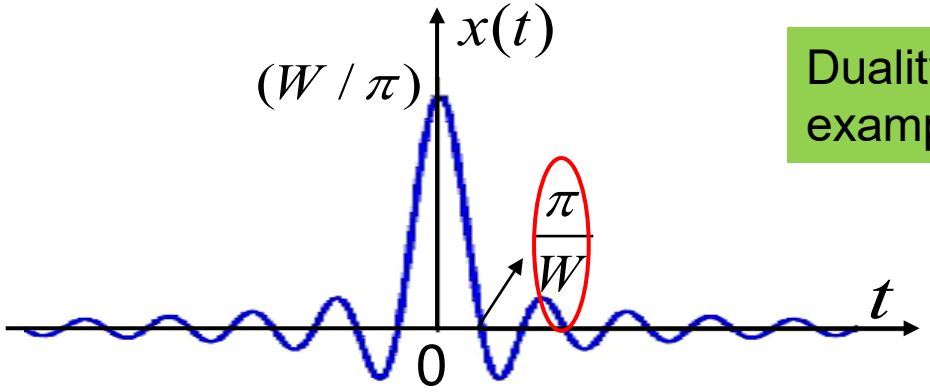
- Example 5: Rectangular frequency signal

$$X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$$

Ideal low-pass filter

$$x(t) = \frac{1}{2\pi}\int_{-W}^{W} e^{j\omega t} d\omega = \frac{\sin Wt}{\pi t} = \frac{W}{\pi}\mathrm{sinc}\left(\frac{Wt}{\pi}\right)$$

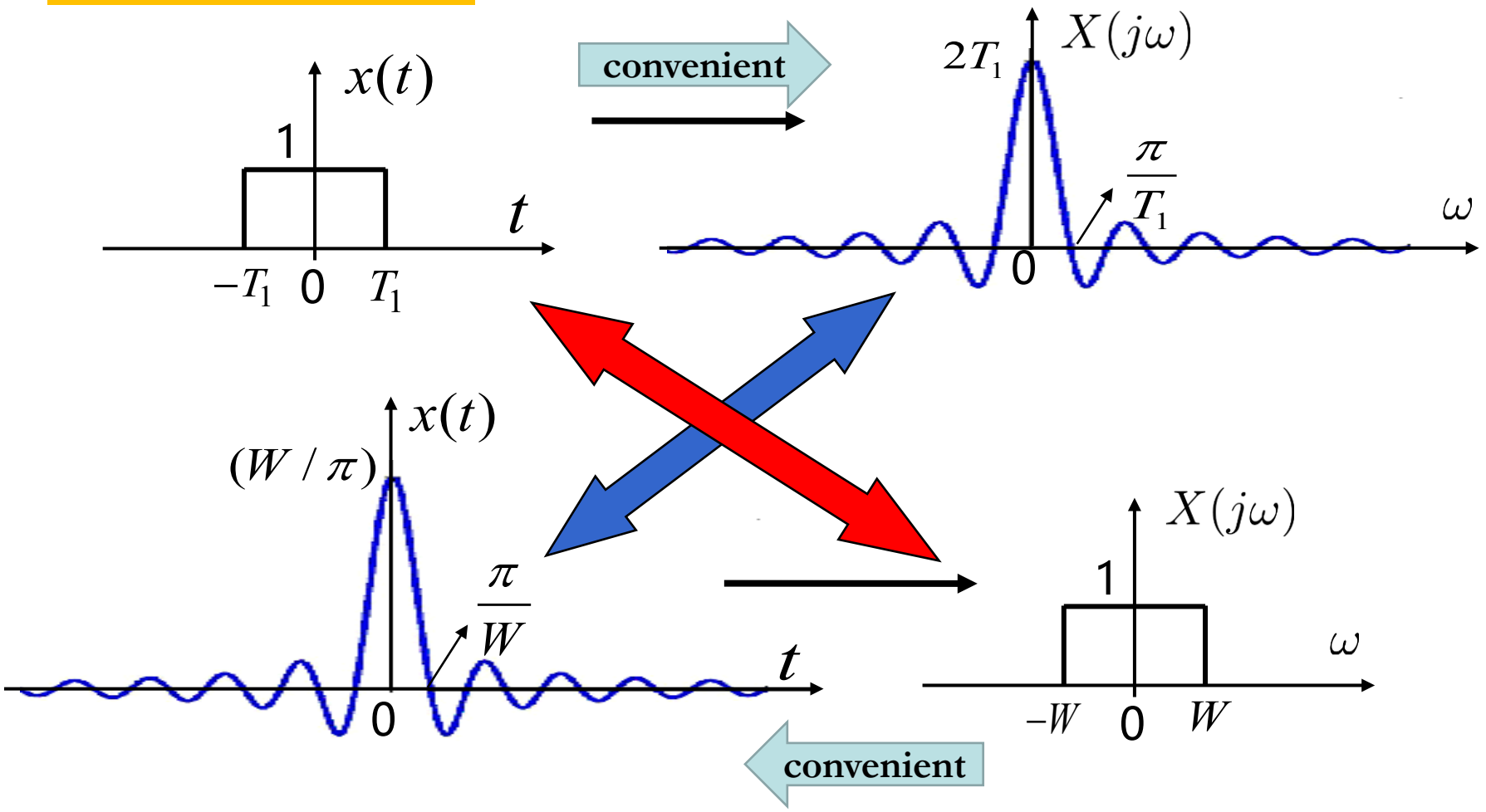




Duality from example 4

Inverse relationship between time and frequency domain

Illustration of duality



- From example 3 and example 4, we can guess as $W \to +\infty$, then $x(t) \to \delta(t)$

Namely, Sinc function goes to $\delta(t)$

- Example 6

$$x(t) = 1 \qquad X(j\omega) = 2\pi\delta(\omega)$$

  - Check

$$\frac{1}{2\pi}\int_{-\infty}^{+\infty} 2\pi\delta(\omega)e^{j\omega t}d\omega = 1$$

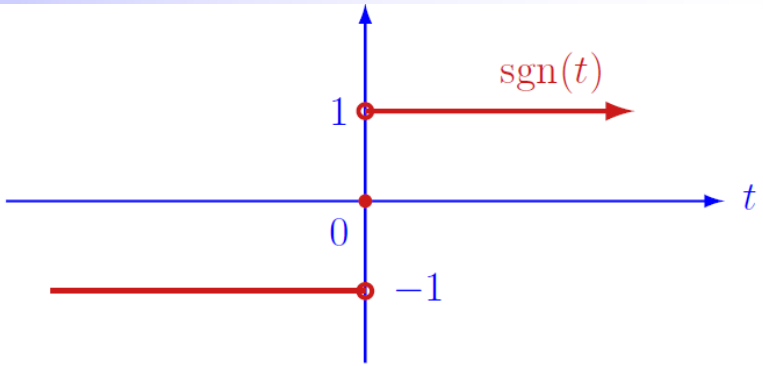
  - From frequency domain back to the time domain is easier

$$x(t) = 1 \overset{\mathcal{F}}{\longleftrightarrow} 2\pi\delta(\omega)$$

convenient

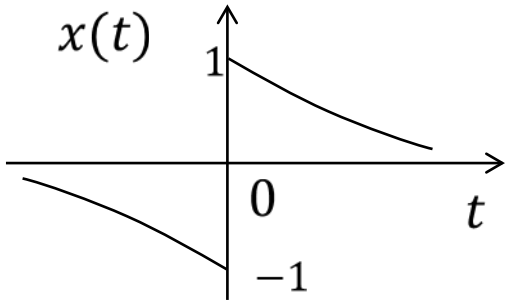
- Example 7: Sign signal

$$\text{sign}(t) = \begin{cases} 1, & t > 0 \\ -1, & t < 0 \end{cases}$$



  - Check the Bilateral Exponential Attenuation Signal

$$x(t) = \begin{cases} e^{-at}u(t), & t > 0 \\ -e^{at}u(-t), & t < 0 \end{cases}, \quad a > 0$$



$$X(j\omega) = \int_0^{+\infty} e^{-at}e^{-j\omega t}dt - \int_{-\infty}^{0} e^{at}e^{-j\omega t}dt$$

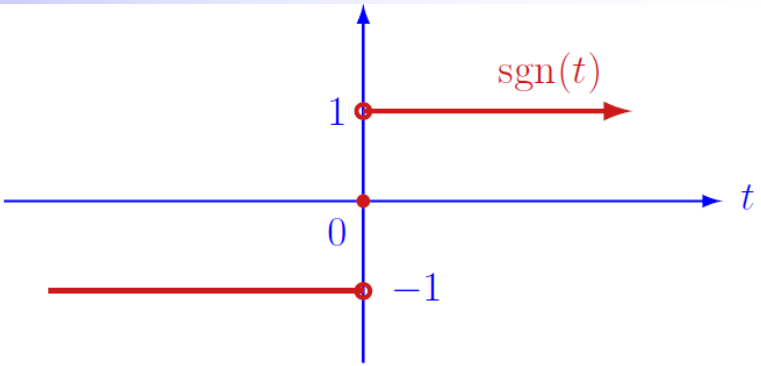
$$= \frac{1}{a+j\omega} - \frac{1}{a-j\omega} = \frac{-2j\omega}{a^2+\omega^2}$$

Could we do?

$$\mathcal{F}[\text{sgn}(t)] = \lim_{a\to 0} X(j\omega) = \frac{2}{j\omega}$$

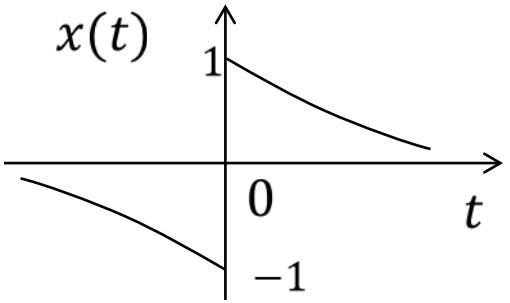
- Example 7: Sign signal

$$\text{sign}(t) = \begin{cases} 1, & t > 0 \\ -1, & t < 0 \end{cases}$$



  - Check the Bilateral Exponential Attenuation Signal

$$x_1(t) = \begin{cases} e^{-at}u(t), & t > 0 \\ -e^{at}u(-t), & t < 0 \end{cases}, \quad a > 0$$

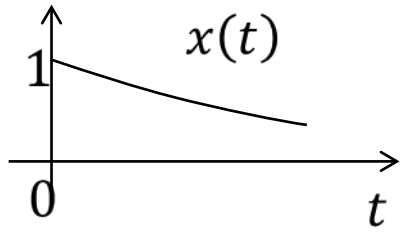


$$X_1(j\omega) = \int_0^{+\infty} e^{-at}e^{-j\omega t}dt + \int_{-\infty}^{0} e^{at}e^{-j\omega t}dt$$

$$= \frac{1}{a+j\omega} - \frac{1}{a-j\omega} = \frac{-2j\omega}{a^2+\omega^2}$$

$\omega \neq 0$ $\mathcal{F}[\text{sgn}(t)] = \lim_{a\to 0} X_1(j\omega) = \frac{2}{j\omega}$ But when $\omega = 0$ $\mathcal{F}[\text{sgn}(t)] = 0$

- Example 8: Unit step function $u(t)$



Check $x(t) = e^{-at}u(t), a > 0$

$$X(j\omega) = \frac{1}{a + j\omega} \xrightarrow{a\to 0} \frac{1}{j\omega}$$

Correct? ? ? Missing $\omega = 0$

$$X(j\omega) = \frac{a}{a^2 + \omega^2} + \frac{-j\omega}{a^2 + \omega^2}$$

Half sign function

$$\frac{j\omega}{a^2 + \omega^2} \xrightarrow{a\to 0} \begin{cases} 0, & \omega = 0 \\ -\frac{1}{j\omega}, & \omega \neq 0 \end{cases}$$

$$\int_{-\infty}^{\infty} \frac{a}{a^2 + \omega^2} d\omega = \int_{-\infty}^{\infty} \frac{1}{1 + \left(\frac{\omega}{a}\right)^2} d\left(\frac{\omega}{a}\right) = \arctan\frac{\omega}{a}\Big|_{-\infty}^{\infty} = \pi$$

$$\frac{a}{a^2 + \omega^2} \xrightarrow{a\to 0} \pi\delta(\omega)$$

$$u(t) \xrightarrow{F} \frac{1}{j\omega} + \pi\delta(\omega)$$

- Alternative way?

The Fourier Transform for Periodic Signals

- Fourier transform is also applicable for periodic signals if impulse function is permitted

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

  - From $\frac{1}{2\pi}\int_{-\infty}^{+\infty} 2\pi\delta(\omega-\omega_0)e^{j\omega t}d\omega = e^{j\omega_0 t}$, we know the Fourier transform for $e^{jk\omega_0 t}$ is $2\pi\delta(\omega-k\omega_0)$
  - So $$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)$$

But previously…..

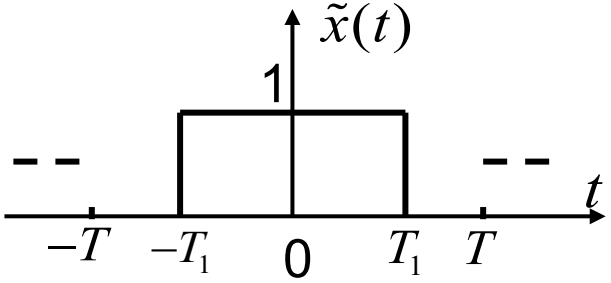
$$a_k = \frac{1}{T}X(j\omega)\Big|_{\omega=k\omega_0}$$

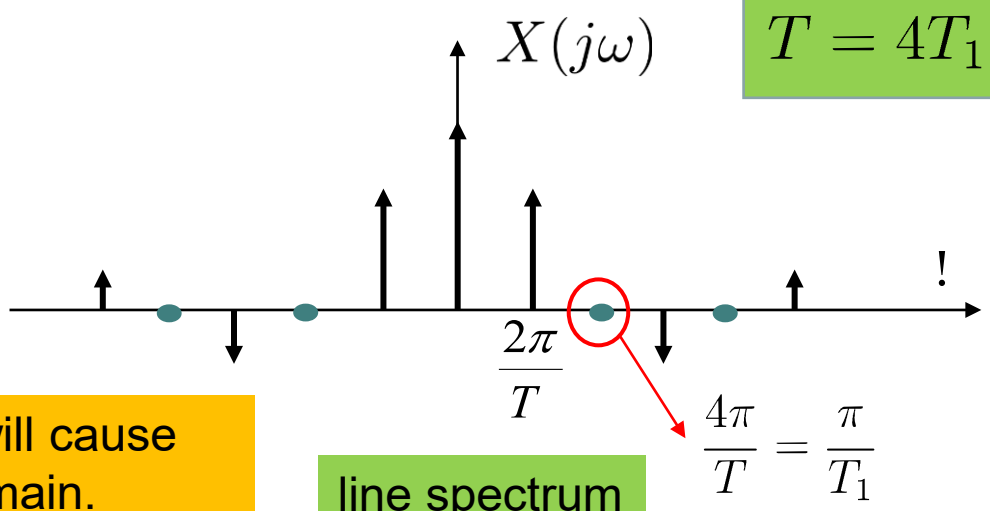
  - The train of impulse occurring at the harmonically related frequencies and the area of the impulse at the kth harmonic frequency $k\omega_0$ is $2\pi a_k$

- Example 1: Periodic square wave

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad \sum_{k=-\infty}^{\infty} a_k \cdot 2\pi\delta(\omega - k\omega_0) = X(j\omega)$$

$$a_k = \frac{\sin k\omega_0 T_1}{\pi k} \quad \Rightarrow \quad X(j\omega) = \sum_{k=-\infty}^{\infty} \frac{2\sin k\omega_0 T_1}{k}\delta(\omega - k\omega_0)$$





$T = 4T_1$

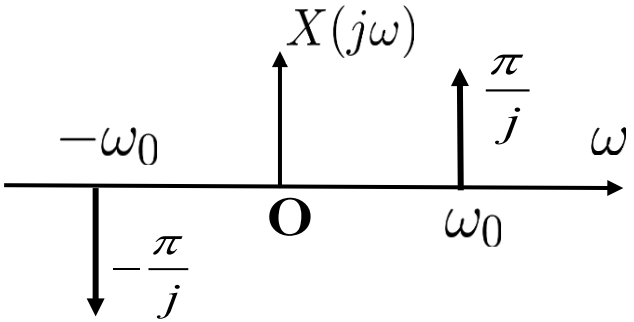
Periodicity in one domain, will cause discretization in another domain.

line spectrum

- Example 2: Sinusoids

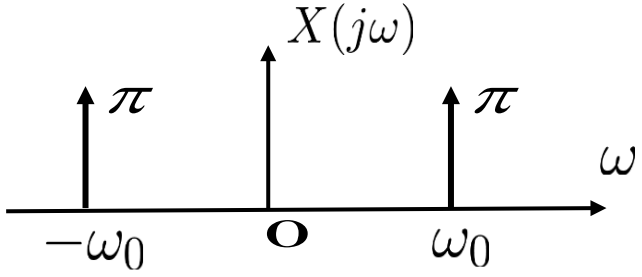
$$x(t) = \sin\omega_0 t = \frac{1}{2j}(e^{j\omega_0 t} - e^{-j\omega_0 t}) \overset{\mathcal{F}}{\longleftrightarrow}$$

$$X(j\omega) = \frac{\pi}{j}[\delta(\omega-\omega_0) - \delta(\omega+\omega_0)]$$

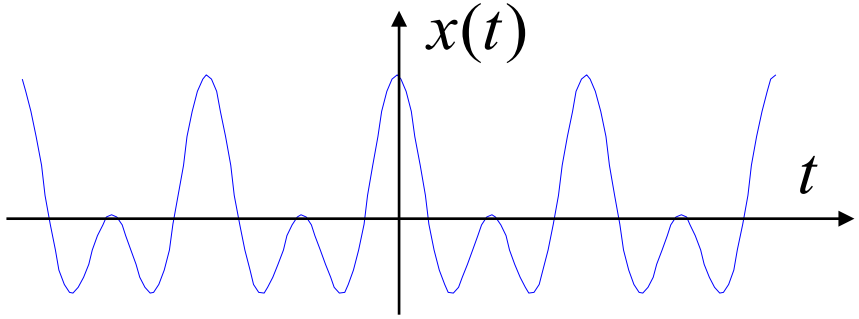


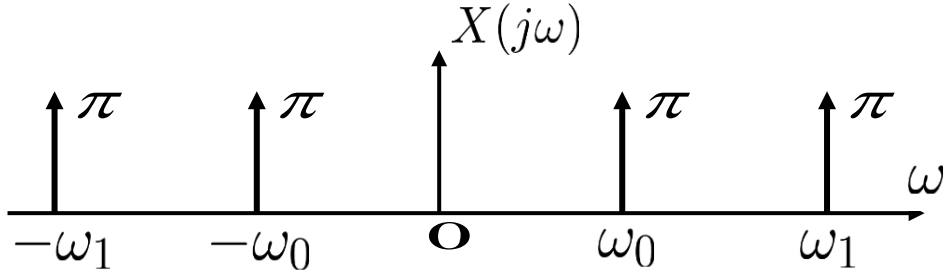
$$x(t) = \cos\omega_0 t = \frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t}) \overset{\mathcal{F}}{\longleftrightarrow}$$

$$X(j\omega) = \pi[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$$



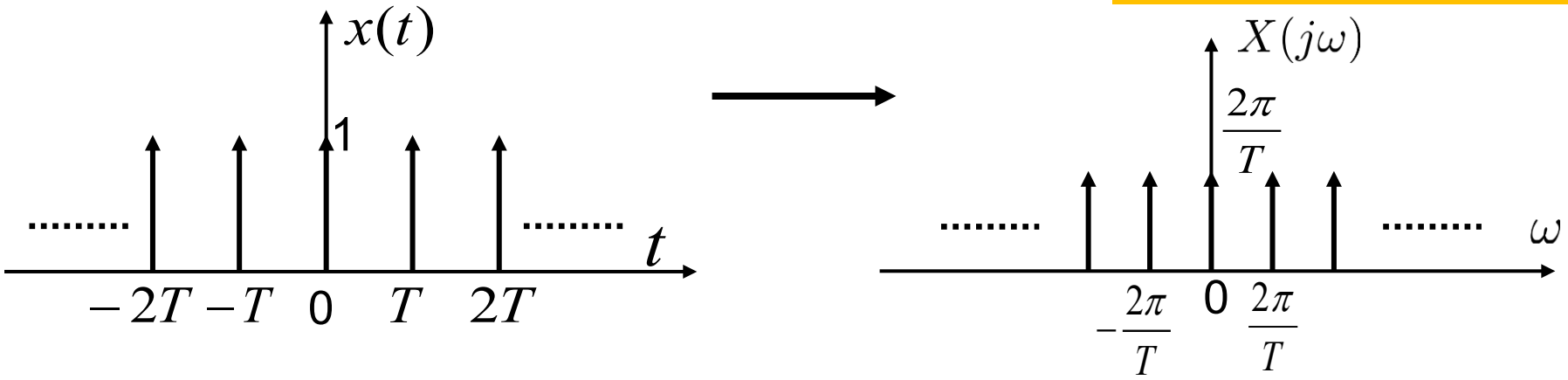
- Example 3: Double sinusoids $x(t) = \cos\omega_0 t + \cos\omega_1 t$





- Example 4: Impulse training
  - From FS $x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT) = \frac{1}{T} \sum_{k=-\infty}^{\infty} e^{jk\omega_0 t}$
  - Hence $X(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$

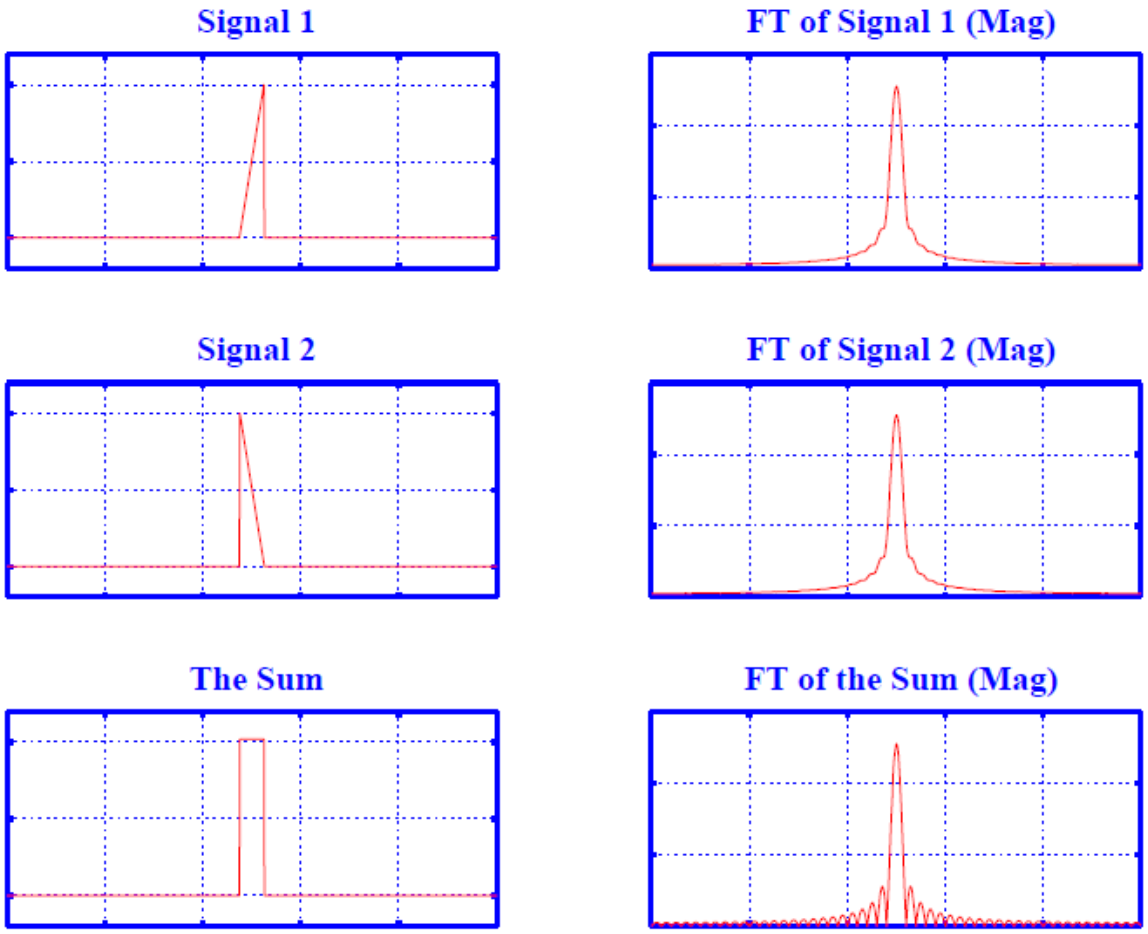
What is the key rationale to have periodicity in time domain?



Inverse relationship between time and frequency domain

Property of CTFT

- Linearity $ax(t) + by(t) \overset{\mathcal{F}}{\longleftrightarrow} aX(j\omega) + bY(j\omega)$



– Example: unit step function

$$u(t) = \frac{1}{2} + \frac{1}{2}\mathrm{sgn}(t)$$

thus

$$\mathcal{F}[u(t)] = \frac{1}{2}\mathcal{F}[1] + \frac{1}{2}\mathcal{F}[\mathrm{sgn}(t)] = \pi\delta(\omega) + \frac{1}{\mathrm{j}\omega}.$$

- Time shifting

$$x(t - t_0) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)e^{-j\omega t_0}$$

Amplitude does not change
Only phase changes $-\omega t_0$

- Time reversal

$$x(-t) \overset{\mathcal{F}}{\longleftrightarrow} X(-j\omega)$$

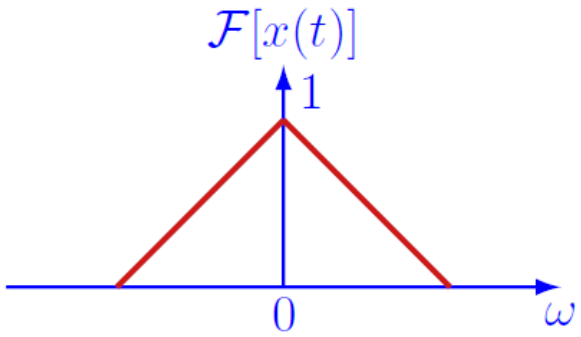
– Check even/odd $x(t)$ for this property

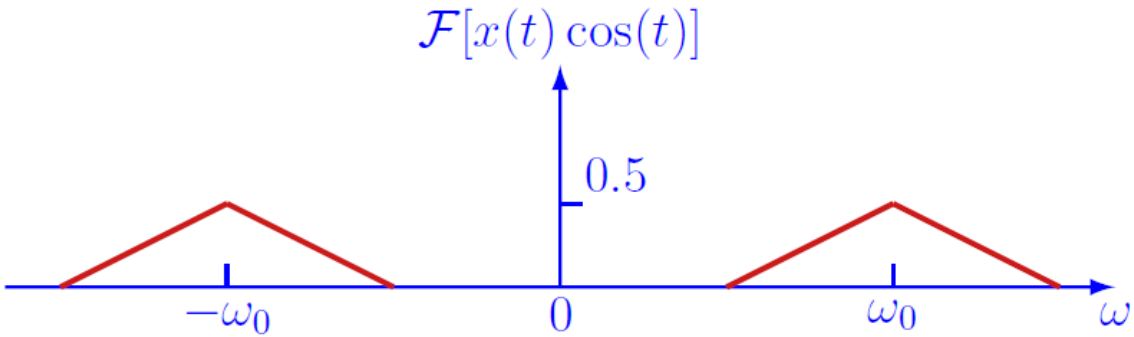
- Frequency shifting

$$x(t)e^{j\omega_0 t} \overset{\mathcal{F}}{\longleftrightarrow} X(j(\omega-\omega_0))$$

  - Hence

$$x(t)\cos(\omega_0 t) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2}\left[X(j(\omega-\omega_0))+X(j(\omega+\omega_0))\right]$$





This result is very important in wireless communications, and will be seen later

- Conjugation $x^*(t) \overset{\mathcal{F}}{\longleftrightarrow} X^*(-j\omega)$
  - Real signal: conjugate symmetry: $X(j\omega) = X^*(-j\omega)$

$$\begin{cases} \Re\{X(j\omega)\} = \Re\{X(-j\omega)\} & \Im\{X(j\omega)\} = -\Im\{X(-j\omega)\} \\ |X(j\omega)| = |X(-j\omega)| & \angle X(j\omega) = -\angle X(-j\omega) \end{cases}$$

  - If $x(t)$ is real and even, so is $X(j\omega)$
  - If $x(t)$ is real and odd, $X(j\omega)$ is purely imaginary and odd.
  - Moreover, rewrite real $x(t)$ as

$$x(t) = x_e(t) + x_o(t)$$

Then $x_e(t) = \frac{1}{2}[x(t) + x(-t)] \overset{\mathcal{F}}{\longleftrightarrow} \Re\{X(j\omega)\}$

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)] \overset{\mathcal{F}}{\longleftrightarrow} j\Im\{X(j\omega)\}$$

This also indicates $\mathcal{F}\{u(t)\} = \frac{1}{j\omega}$ is not correct

- Time domain Differentiation: $\frac{dx(t)}{dt} \overset{\mathcal{F}}{\longleftrightarrow} j\omega X(j\omega)$ For LCCDE later

- Time domain integration: $\int_{-\infty}^{t} x(\tau)d\tau \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)$

$$\mathcal{F}\left[\int_{-\infty}^{t} x(\tau)\mathrm{d}\tau\right] = \int_{-\infty}^{\infty}\left[\int_{-\infty}^{t} x(\tau)\mathrm{d}\tau\right]\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$= \int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} x(\tau)u(t-\tau)\mathrm{d}\tau\right]\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

Will see a more general convolution property later

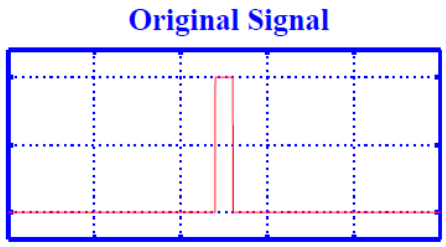
$$= \int_{-\infty}^{\infty} x(\tau)\left[\int_{-\infty}^{\infty} u(t-\tau)\mathrm{e}^{-\mathrm{j}\omega t}dt\right]d\tau$$

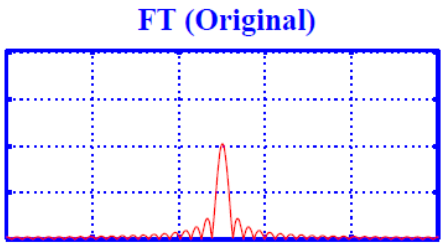
$$(\text{Time Shifting \& } \mathcal{F}[u(t)]) = \int_{-\infty}^{\infty} x(\tau)\left[\pi\delta(\omega) + \frac{1}{\mathrm{j}\omega}\right]\mathrm{e}^{-\mathrm{j}\omega\tau}d\tau$$

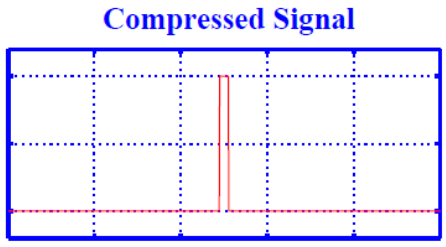
$$= \frac{1}{\mathrm{j}\omega}X(\mathrm{j}\omega) + \pi X(0)\delta(\omega).$$

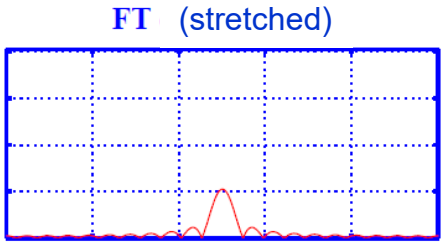
- Time and frequency scaling $x(at) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$
  - Compressed in time
    Stretched in frequency
    $|a| > 1$
  - Stretched in time
    Compressed in frequency
    $|a| < 1$
  - Reverse a signal in time
    also reverses its FT
    $a < 0$

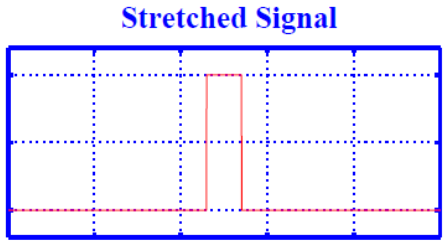
How about FS?

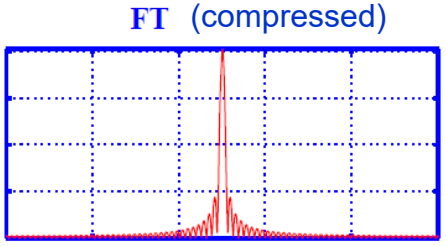












Inverse relationship between time and frequency domain

- Parseval's relation:

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

Energy density

Total energy

- Duality

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

two differences

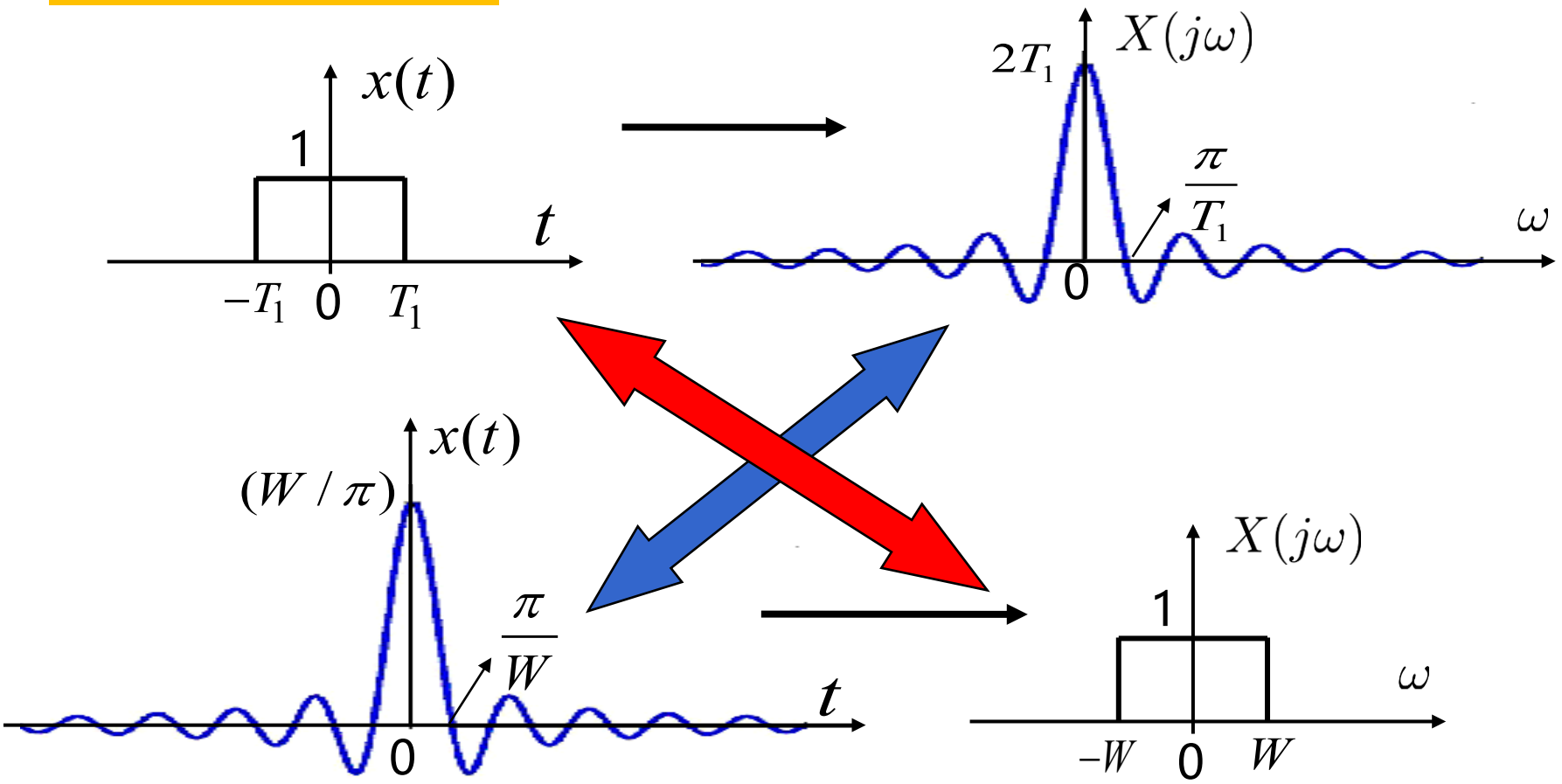
注意(jt)不是把w换成jt
而是把w换成t

  - In fact, there is $X(jt) \xleftrightarrow{\mathcal{F}} 2\pi x(-\omega)$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\mathrm{j}\omega) \mathrm{e}^{\mathrm{j}\omega t} \mathrm{d}\omega \Longrightarrow x(-t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\mathrm{j}\omega) \mathrm{e}^{-\mathrm{j}\omega t} \mathrm{d}\omega$$

$$\Longrightarrow 2\pi x(-\omega) = \int_{-\infty}^{+\infty} X(\mathrm{j}t) \mathrm{e}^{-\mathrm{j}\omega t} \mathrm{d}t.$$

Illustration of duality



– Example $\mathcal{F}[\delta(t)] = 1$  $\mathcal{F}[1] = 2\pi\delta(-\omega) = 2\pi\delta(\omega)$

– Example $e^{-|t|} = x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega) = \dfrac{2}{1+\omega^2}$



$$\frac{2}{1+t^2} = X(jt) \overset{\mathcal{F}}{\longleftrightarrow} 2\pi x(-\omega) = 2\pi e^{-|\omega|}$$

– Other properties

Frequency differentiation $\quad -jtx(t) \overset{\mathcal{F}}{\longleftrightarrow} \dfrac{dX(j\omega)}{d\omega}$

Frequency integration $\quad -\dfrac{1}{jt}x(t) + \pi x(0)\delta(t) \overset{\mathcal{F}}{\longleftrightarrow} \displaystyle\int_{-\infty}^{\omega} X(j\tau)d\tau$

The Convolution Property

- Almost the most important property of this course

$$h(t) * x(t) \overset{\mathcal{F}}{\longleftrightarrow} H(j\omega)X(j\omega)$$

  - We can modify $x(t)$ through modifying $X(j\omega)$ by $H(j\omega)$

$$\begin{aligned}
\mathcal{F}\left[x(t) * h(t)\right] &= \int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} x(\tau)h(t-\tau)\mathrm{d}\tau\right]\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t \\
&= \int_{-\infty}^{\infty} x(\tau)\left[\int_{-\infty}^{\infty} h(t-\tau)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t\right]\mathrm{d}\tau \\
&= \int_{-\infty}^{\infty} x(\tau)H(\mathrm{j}\omega)\mathrm{e}^{-\mathrm{j}\omega\tau}\mathrm{d}\tau \\
&= X(\mathrm{j}\omega)H(\mathrm{j}\omega).
\end{aligned}$$

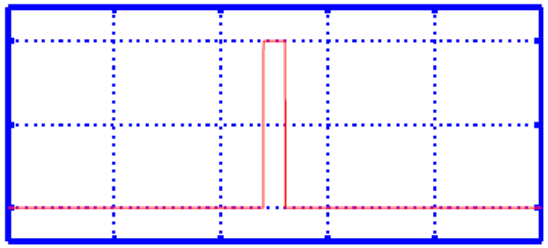
Remember our LTI system

May also check the wordy proof from the limitation of FS of periodic signal in book

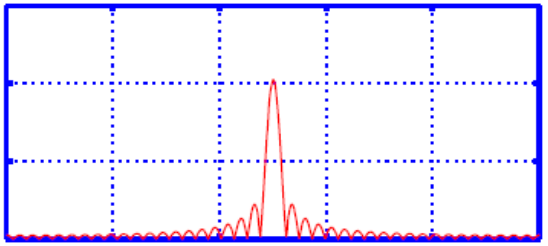
- Example

The second signal may not be h(t) and can be any signal!

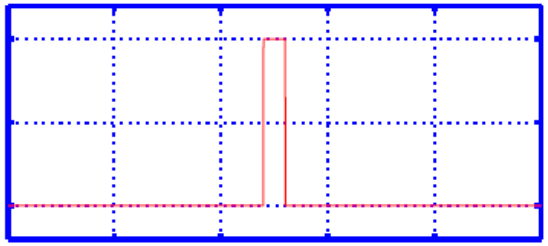
Signal 1



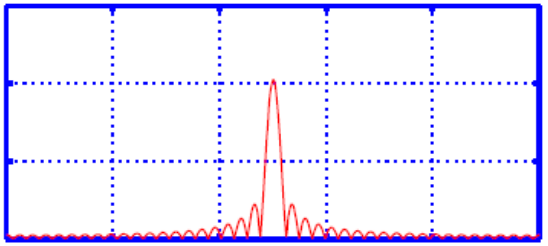
FT (Signal 1)



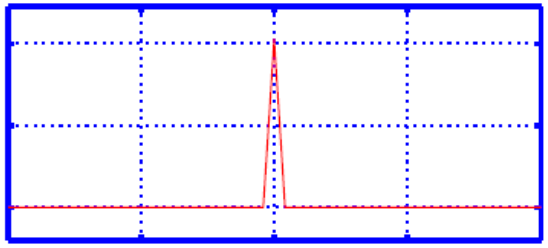
Signal 2



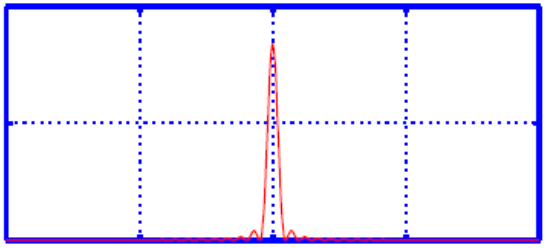
FT (Signal 2)



The Convolution of Two Signals



FT (Convolution)



- Explanation: $x(t)$ can be view as the linear combination of infinite number of complex exponentials

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$$

- After passing through the LTI $h(t)$, each exponentials $e^{j\omega t}$ would have a response $H(j\omega)e^{j\omega t}$, where

$$H(j\omega) = \int_{-\infty}^{+\infty} h(t)e^{-j\omega t}dt$$

- The out of the LTI would be the linear combination of all these component and hence

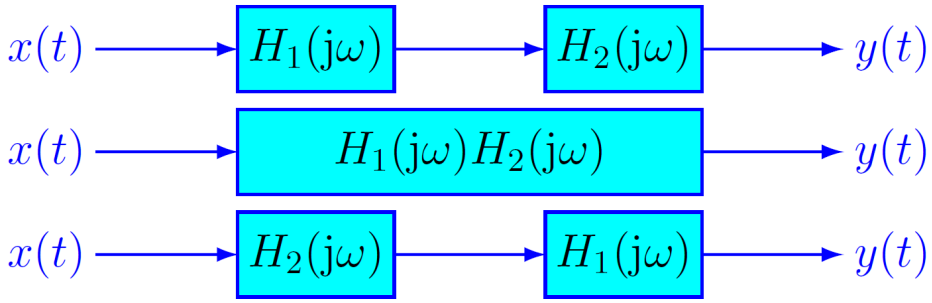
$$Y(j\omega) = X(j\omega)H(j\omega)$$

LTI convert the time domain convolution into the frequency domain multiplication

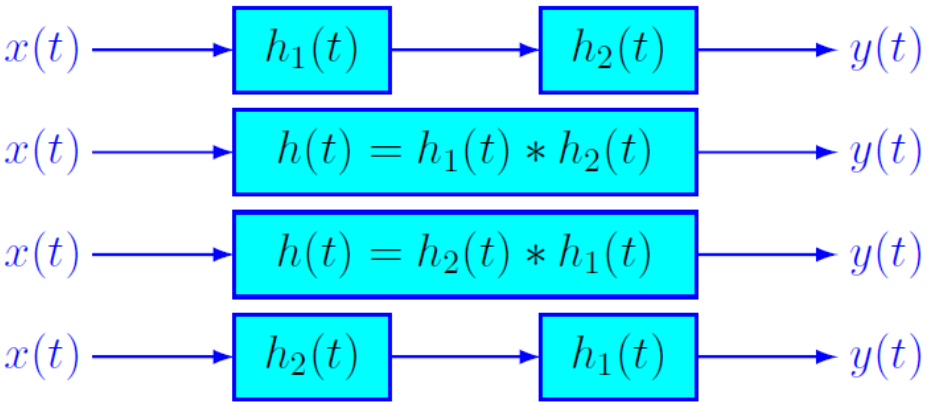
The rationale is that the complex exponentials is the eigen-function for ANY LTI

- Fourier transform provides us a frequency-domain analysis method to study the signals and systems.

Frequency-domain analysis



Time-domain analysis



When system frequency response exists （remember the two sets of conditions）, the two analysis are the same.

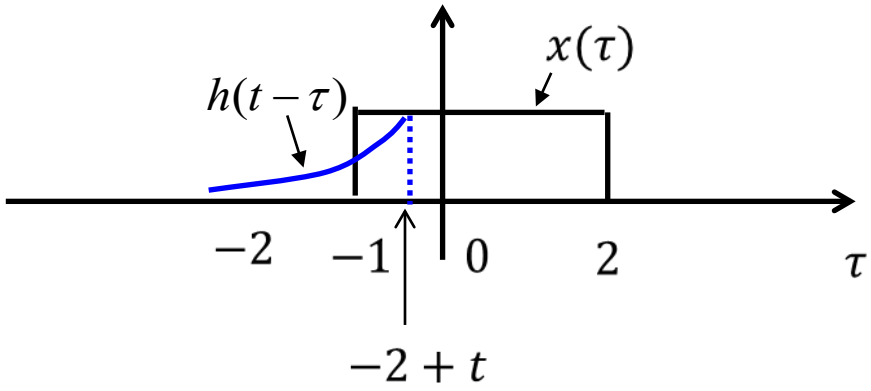
Otherwise, check the Laplace transform

Signals Pass LTI

- Clearly， we can design LTI $H(j\omega)$, to modify $X(j\omega)$

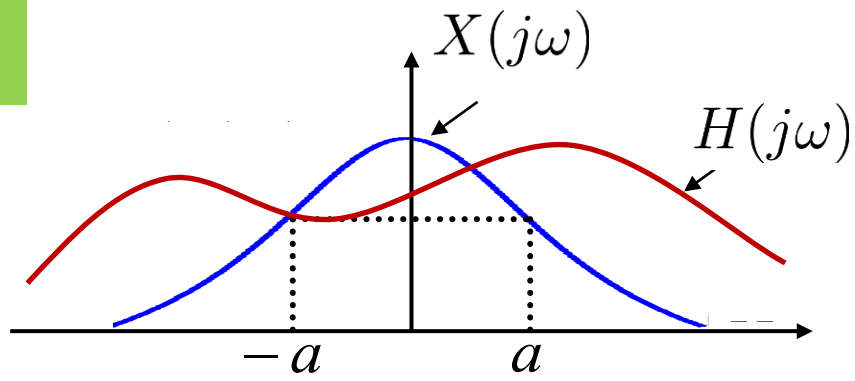
Time domain convolution

$$h(t) * x(t)$$



Frequency domain multiplication
Frequency selectivity

$$Y(j\omega) = X(j\omega)H(j\omega)$$



- Filtering
  - LPF, HPF, BPF

- Frequency domain analysis
  - From $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$
  - Compute $H(j\omega)$
  - Compute $Y(j\omega) = X(j\omega)H(j\omega)$
  - Get the inverse $Y(j\omega) \overset{\mathcal{F}^{-1}}{\longleftrightarrow} y(t)$
  - Frequency domain analysis can avoid the complicated convolution and present clearer physical meaning.
  - Possibly to solve the LCCDE in a easier way.
- Example: delay system $h(t) = \delta(t - t_0)$

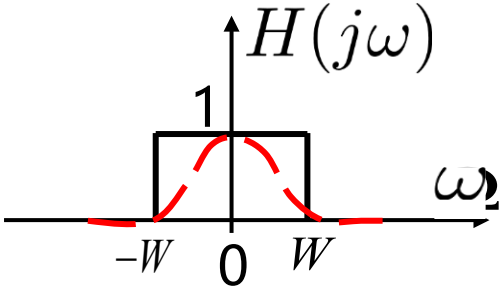
$$H(j\omega) = e^{-j\omega t_0} \Rightarrow Y(j\omega) = e^{-j\omega t_0} X(j\omega)$$

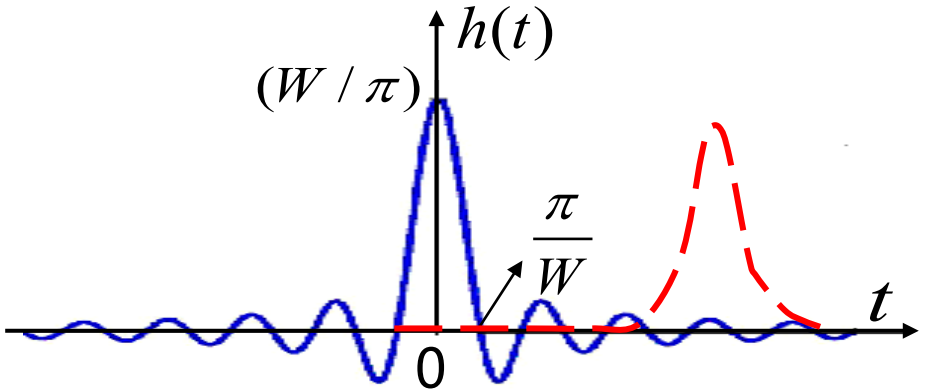
$$\Rightarrow y(t) = x(t - t_0)$$

- Example: Ideal low pass filter

$$H(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$$

$$h(t) = \frac{1}{2\pi}\int_{-W}^{W} e^{j\omega t} d\omega = \frac{\sin Wt}{\pi t} = \frac{W}{\pi}\text{sinc}\left(\frac{Wt}{\pi}\right)$$





The idea LPF is not causal and cannot be realized. The practical design must take on certain trade-off (see red curve)

- Example: Find the output of the following system with

$$h(t) = e^{-at}u(t), \quad a > 0 \qquad x(t) = e^{-bt}u(t), \quad b > 0$$

$$X(j\omega) = \frac{1}{b + j\omega}, \; H(j\omega) = \frac{1}{a + j\omega}, \; Y(j\omega) = \frac{1}{(a + j\omega)(b + j\omega)}$$

  - When $b \neq a$: partial fraction expansion

$$Y(j\omega) = \frac{A}{a + j\omega} + \frac{B}{b + j\omega} \qquad \begin{cases} A = -B = \dfrac{1}{b - a} \\ y(t) = Ae^{-at}u(t) + Be^{-bt}u(t) \end{cases}$$

  - When $b = a$: check table

$$Y(j\omega) = \frac{1}{(a + j\omega)^2} \quad \Rightarrow \quad y(t) = te^{-at}u(t) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{(a + j\omega)^2}$$

How to prove smartly?

- Example: Find the output of the following system with

$$h(t) = e^{-at}u(t), \quad a > 0 \qquad\qquad x(t) = e^{-bt}u(t), \quad b > 0$$

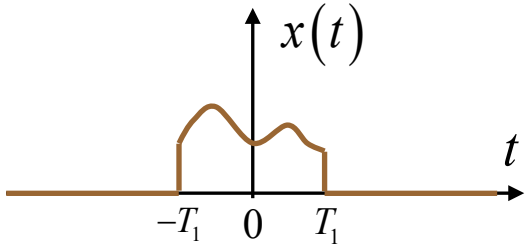
$$y(t) = x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$$

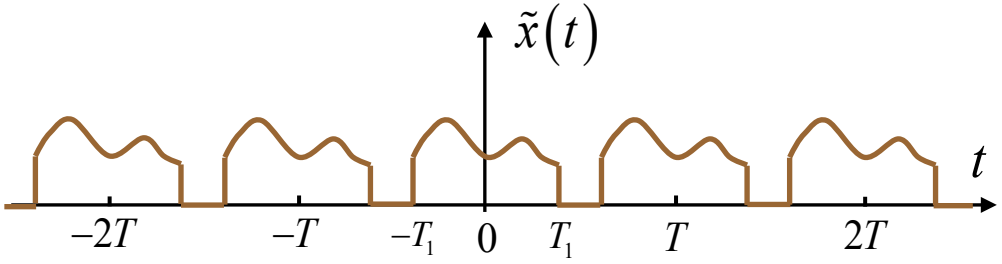
$$= \int_0^t e^{-b\tau}e^{a(\tau-t)}d\tau \cdot u(t)$$

$$y(t) = \begin{cases} \int_0^t e^{-at}d\tau u(t) = e^{-at}\int_0^t d\tau u(t) = te^{-at}u(t), & a = b \\ e^{-at}\int_0^t e^{(a-b)\tau}d\tau u(t) = \frac{e^{-at}}{a-b}\cdot e^{(a-b)\tau}\Big|_0^t \cdot u(t), & a \neq b \end{cases}$$

$$= \begin{cases} te^{-at}u(t), & a = b \\ \frac{e^{-at}}{a-b}\left[e^{(a-b)t} - 1\right]u(t) = \frac{1}{a-b}\left\{e^{-bt} - e^{-at}\right\}u(t), & a \neq b \end{cases}$$

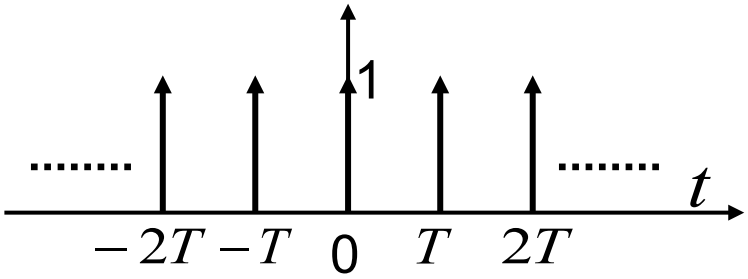
- Example:

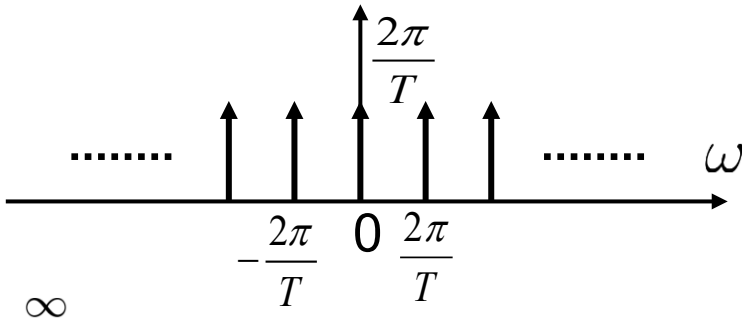




$$h(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$H(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$$





$$\tilde{x}(t) = x(t) * h(t) \quad \Rightarrow \quad \tilde{x}(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} X(jk\omega_0)\delta(\omega - k\omega_0)$$

Sampling in frequency domain

- Consider the impulse response $h(t) = \dfrac{\sin 2\pi t}{\pi t}$ and the input

$$x_1(t) = \cos\frac{\pi}{2}t, \quad x_2(t) = \frac{1}{3} + \frac{2}{3}\left(\cos\frac{2\pi}{3}t + \cos\frac{4\pi}{3}t + \cos 3\pi t\right)$$

determine the output $y_1(t), \quad y_2(t)$

The frequency impulse response is given by

$$H(j\omega) = \begin{cases} 1, & |\omega| < 2\pi \\ 0, & |\omega| > 2\pi \end{cases}$$

LPF

Only the input with frequency lower 2π can pass. Hence,

$$y_1(t) = x_1(t), \quad y_2(t) = \frac{1}{3} + \frac{2}{3}\left(\cos\frac{2\pi}{3}t + \cos\frac{4\pi}{3}t\right)$$

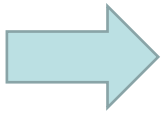
The Multiplication Property

- Application in signal modulation

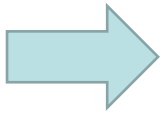
$$x_1(t)x_2(t) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2\pi} X_1(j\omega) * X_2(j\omega)$$

  - Prove from duality

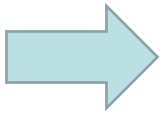
$$X_1(jt) \overset{\mathcal{F}}{\longleftrightarrow} 2\pi x_1(-\omega) \quad X_2(jt) \overset{\mathcal{F}}{\longleftrightarrow} 2\pi x_2(-\omega)$$



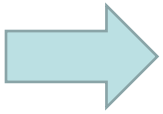
$$X_1(jt) * X_2(jt) \overset{\mathcal{F}}{\longleftrightarrow} 4\pi^2 x_1(-\omega)x_2(-\omega)$$



$$X_1(-jt) * X_2(-jt) \overset{\mathcal{F}}{\longleftrightarrow} 4\pi^2 x_1(\omega)x_2(\omega)$$



$$4\pi^2 x_1(t)x_2(t) \overset{\mathcal{F}}{\longleftrightarrow} 2\pi X_1(j\omega) * X_2(j\omega)$$



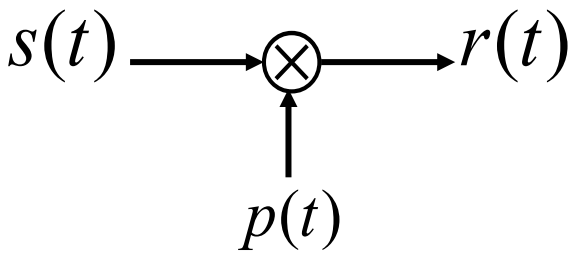
$$x_1(t)x_2(t) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2\pi} X_1(j\omega) * X_2(j\omega)$$

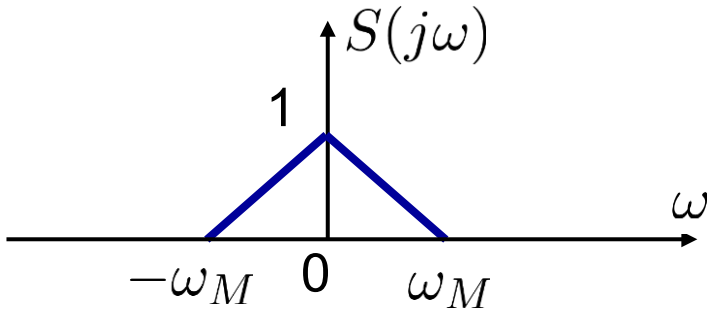
The Multiplication Property

- Prove from direct computation

$$\begin{aligned}
\int_{-\infty}^{\infty} x_1(t)\cdot x_2(t)e^{-j\omega t}dt &= \int_{-\infty}^{\infty}\frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(j\Omega)e^{j\Omega t}d\Omega\cdot x_2(t)e^{-j\omega t}dt \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(j\Omega)\int_{-\infty}^{\infty} x_2(t)e^{-j(\omega-\Omega)t}dtd\Omega \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(j\Omega)X_2[j(\omega-\Omega)]d\Omega \\
&= \frac{1}{2\pi}X_1(j\omega) * X_2(j\omega)
\end{aligned}$$

- In practical application, the multiplication is often used for modulation, frequency conversion and sampling.
  - Time domain multiplication can be viewed as using one signal to control the amplitude of another, which is called as amplitude modulation.
  - One is called modulating signal and another is carrier signal
  - Multiplication property is also called as modulation property

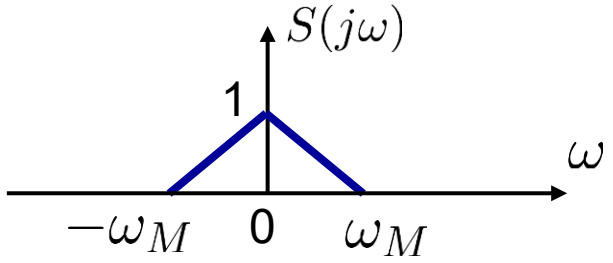




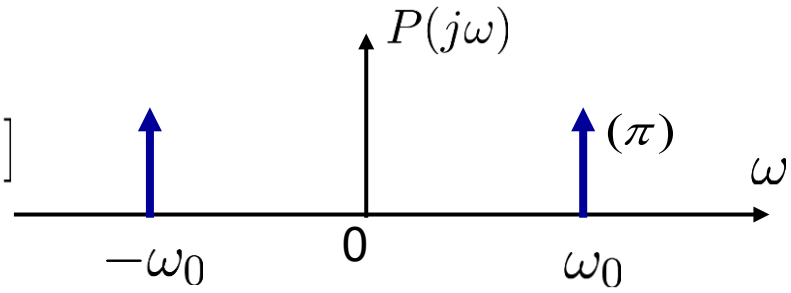
- Example: $p(t) = \cos \omega_0 t$

$$r(t) = s(t)p(t)$$

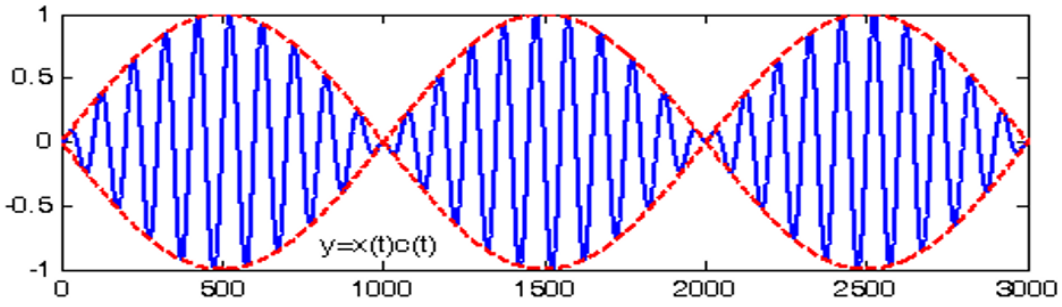
ω_0 is not the fundamental frequency for periodic signal

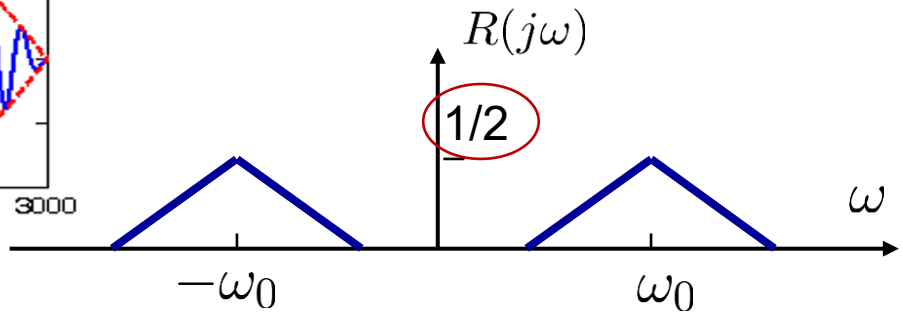


$$P(j\omega) = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$



$$R(j\omega) = \frac{1}{2}[S(j(\omega - \omega_0)) + S(j(\omega + \omega_0))]$$



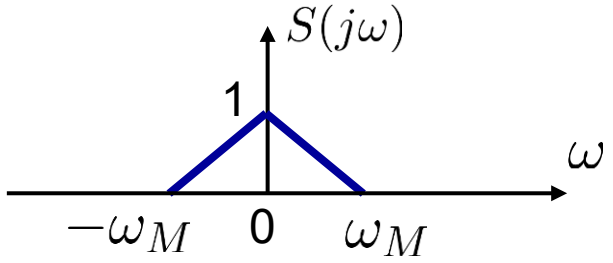


Modulation with sinusoids is just to move the spectrum to the carrier center frequency

- Example: $p(t) = \cos \omega_0 t$

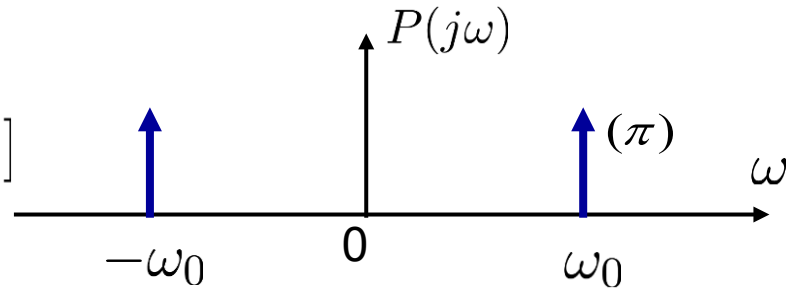
$$r(t) = s(t)p(t)$$

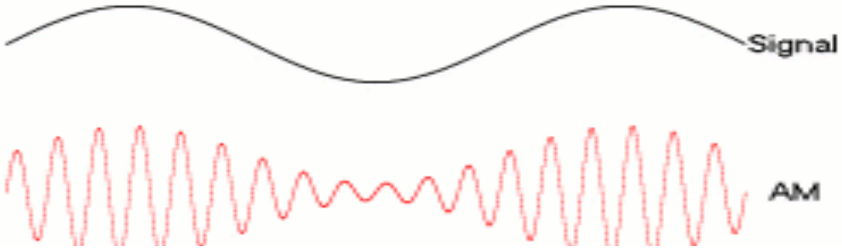
ω_0 is not the fundamental frequency for periodic signal

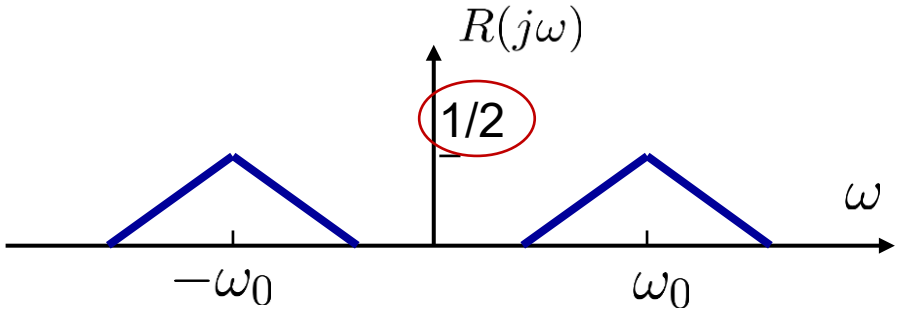


$$P(j\omega) = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$R(j\omega) = \frac{1}{2}[S(j(\omega - \omega_0)) + S(j(\omega + \omega_0))]$$







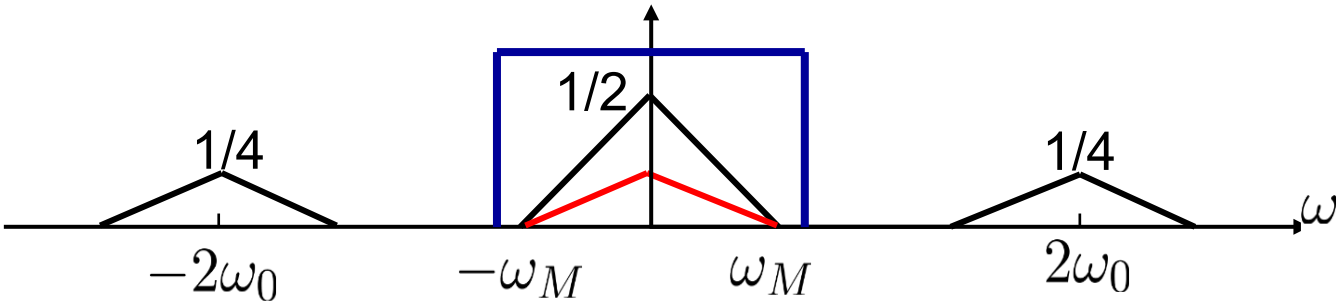
Modulation with sinusoids is just to move the spectrum to the carrier center frequency

- Demodulation $s(t)$

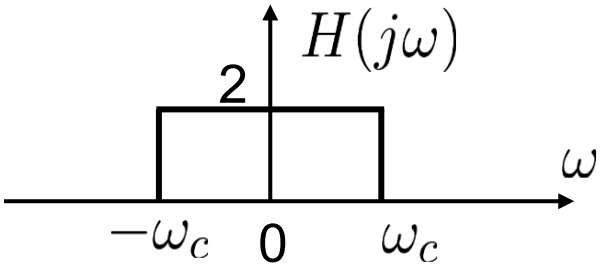
What if the carrier signal is $e^{j\omega_0 t}$?

$$r(t)\cos\omega_0 t \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2}[R(j(\omega-\omega_0))+R(j(\omega+\omega_0))]$$

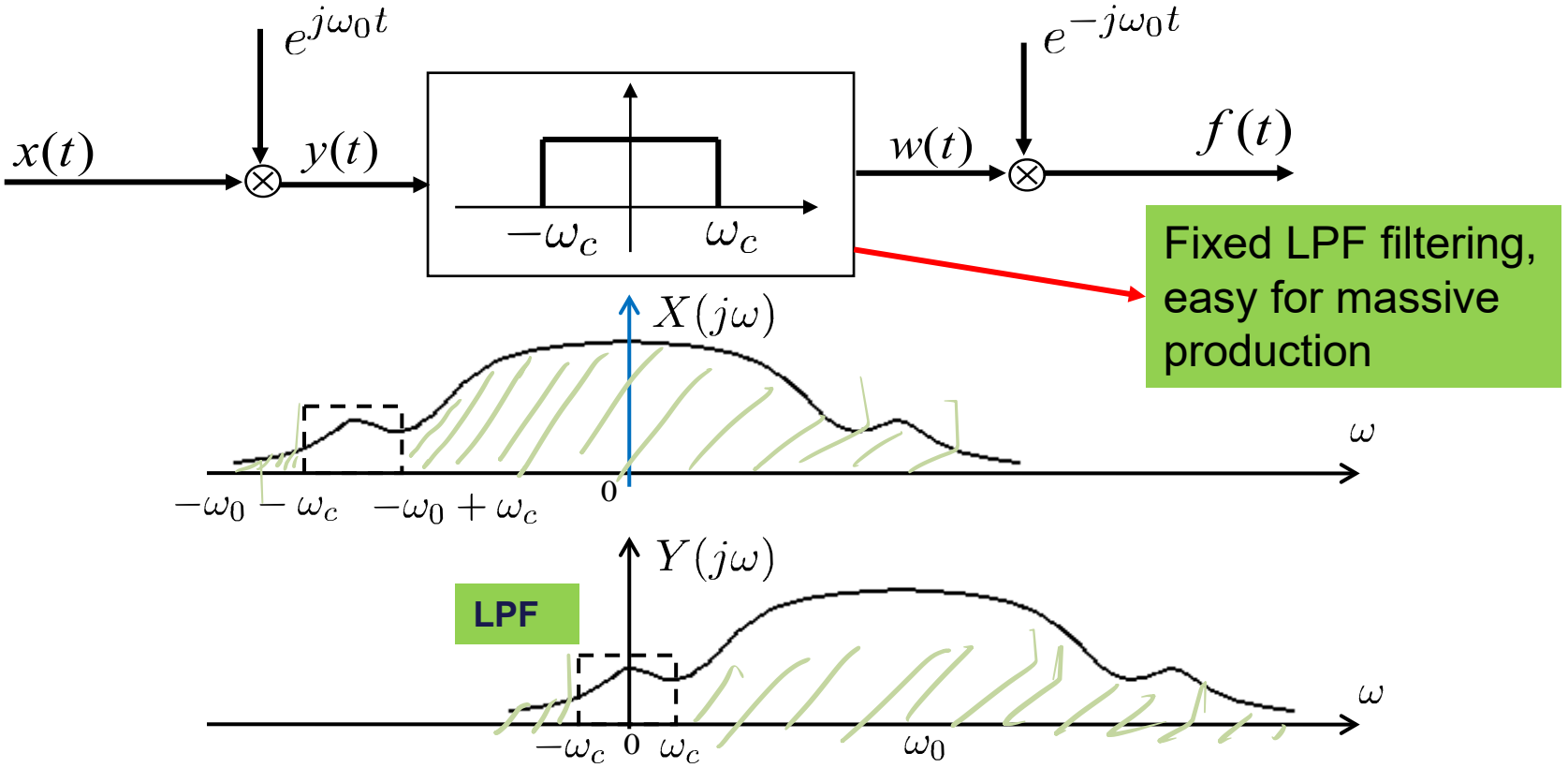
$$= \frac{1}{2}S(j\omega)+\frac{1}{4}[S(j(\omega-2\omega_0))+S(j(\omega+2\omega_0))]$$

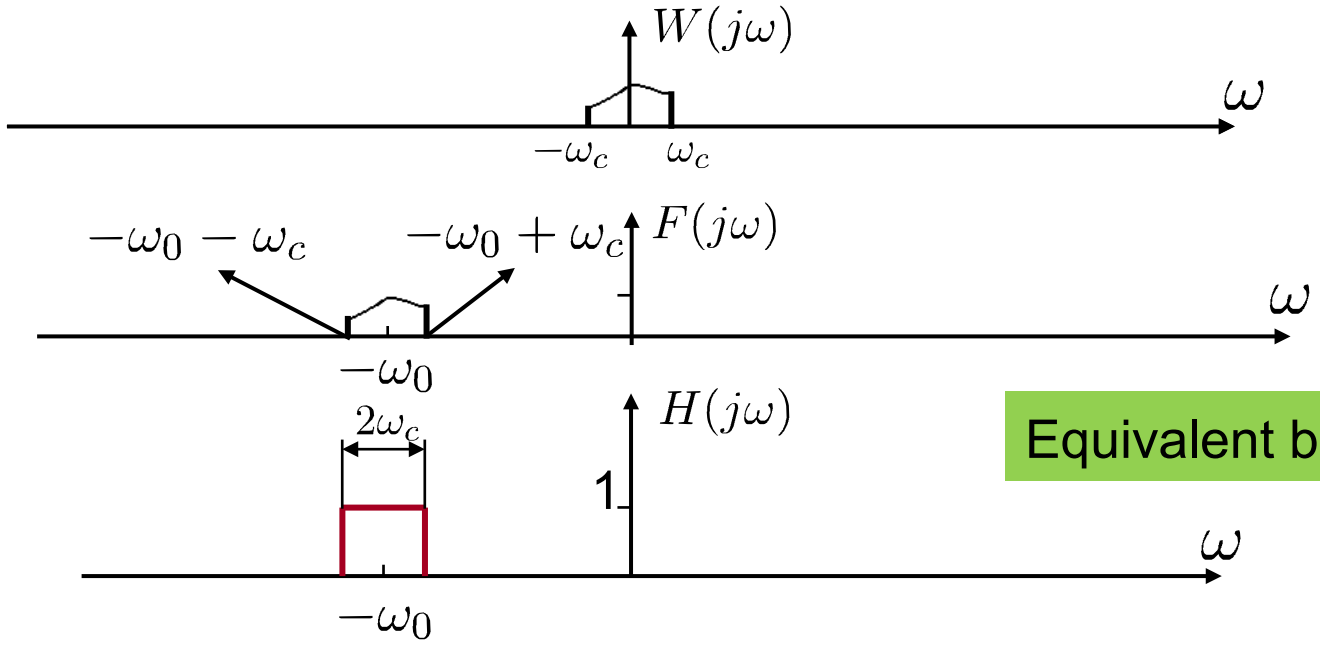


We need one more LTI system that have response $H(j\omega)$ to recover $s(t)$ from $r(t)$ as long as $\omega_M < \omega_c < 2\omega_0 - \omega_M$



- Example： filter with varying center frequency
  - It is hard to directly varying the frequency center of filtering
  - Use filter with fixed center frequency and the shifting property





Equivalent bandpass filter

The above operation is an ideal bandpass filter with center frequency $-\omega_0$ and bandwidth $2\omega_c$. One can simply change ω_0 to change the center frequency.

System Characterized by Linear Constant-Coefficient Differential Equations.

- From previous chapter, we know that a lot of LTI systems can be described as LCCDE

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

  - Previous forced response method, we utilized the property that the input $x(t) = e^{j\omega_0 t}$ gives the output $y(t) = H(j\omega_0)e^{j\omega_0 t}$
  - Substituting these two expressions into LCCDE and we solve for expression of $y(t)$
  - However:
    - The forced response method to solve LCCDE is too complicated
    - We do not get the impulse response $h(t)$ and hence it is not general

System Characterized by Linear Constant-Coefficient Differential Equations.

- Alternative approach takes FT at both sides and obtain

$$\sum_{k=0}^{N} a_k (j\omega)^k Y(j\omega) = \sum_{k=0}^{M} b_k (j\omega)^k X(j\omega)$$

$$\Rightarrow \quad Y(j\omega) = \frac{\sum_{k=0}^{M} b_k (j\omega)^k}{\sum_{k=0}^{N} a_k (j\omega)^k} X(j\omega)$$

  - Hence, the frequency response is

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum_{k=0}^{M} b_k (j\omega)^k}{\sum_{k=0}^{N} a_k (j\omega)^k}$$

  - Hence, $H(j\omega)$ of and LCCDE is a rational function and when we want to get $h(t)$, we normally get $H(j\omega)$ and perform the inverse FT.
  - Then, solving LCCDE becomes the algebra

- Example: Frequency response and impulse response of the following stable LTI system

$$\frac{d^2y(t)}{dt^2}+6\frac{dy(t)}{dt}+8y(t)=\frac{dx(t)}{dt}+3x(t)$$

$$H(j\omega)=\frac{j\omega+3}{(j\omega)^2+6(j\omega)+8}=\frac{j\omega+3}{(j\omega+4)(j\omega+2)}$$

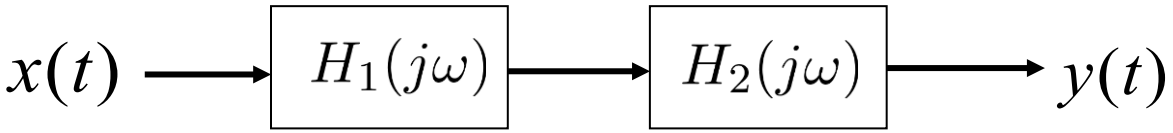
$$=\frac{1}{2}\left[\frac{1}{(j\omega+2)}+\frac{1}{(j\omega+4)}\right]$$

$$h(t)=\frac{1}{2}[e^{-2t}+e^{-4t}]u(t)$$

Hence, the impulse response of LCCDE can be easily obtained from frequency response

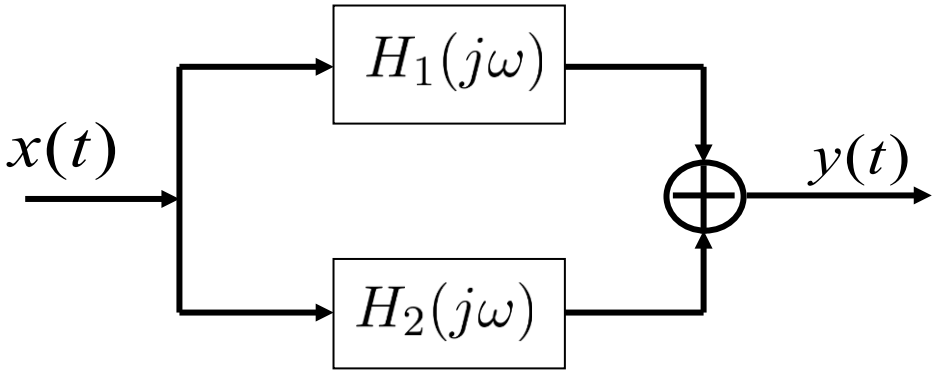
Interconnection of LTI Systems

Cascade



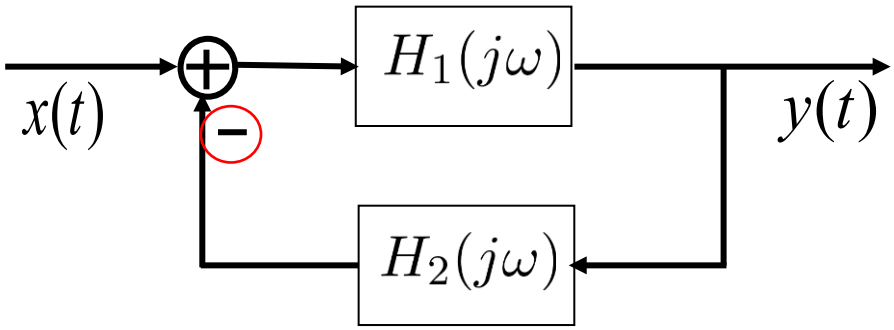
$$H(j\omega) = H_1(j\omega)H_2(j\omega)$$

Parallel



$$H(j\omega) = H_1(j\omega) + H_2(j\omega)$$

Feedback



$$Y(j\omega) = [X(j\omega) - Y(j\omega)H_2(j\omega)]H_1(j\omega)$$

$$X(j\omega)H_1(j\omega) = Y(j\omega)[1 + H_1(j\omega)H_2(j\omega)]$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{H_1(j\omega)}{1 + H_1(j\omega)H_2(j\omega)}$$

Summary

- With CTFT, we develop the way to represent the continuous time signal (periodic, aperiodic) as the linear combination of the complex exponentials.
- With the properties of FT, we enclose the relationship between time-domain and frequency-domain. Specifically, the convolution is the foundation for the frequency analysis of LTI system, while the multiplication is the foundation of communications signal and modulation/demodulation.
- We develop the frequency domain analysis
- For LCCDE described LTI system, we obtain the frequency response in an easier way, and then get the impulse response
- We build the interconnection of LTI system, and derive their frequency response

Homework #4

- 4.11, 4.13, 4.15, 4.17, 4.19, 4.21(a)(c)(f), 4.25, 4.30, 4.39